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Sedimentary Origin of the Mn-Fe Ore of Um Bogma, Southwest Sinai: Geochemical and Paleomagnetic Evidence

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Abstract

Paleomagnetic, mineralogical, and geochemical investigations have been carried out to estimate the origin and age of the Mn-Fe ore of Um Bogma Formation of the early Carboniferous. For this purpose, oriented samples were collected from 26 sites in the Mn-Fe ore and host rocks of Um Bogma Formation and the overlying sandstone of Abu Thora Formation. The paleomagnetic pole, determined from the characteristic remanent magnetization of the Mn-Fe ore from 20 sites and the hosting rocks of Um Bogma Formation, is located at 22° S, 97° E (Dp = 3.9°, Dm = 7.4°). This paleomagnetic pole position falls close to the Carboniferous poles of Africa, suggesting that the Mn-Fe ore and associated rocks of Um Bogma Formation were deposited during the Carboniferous. Furthermore, the paleolatitude determined for these rocks, based on the mean paleomagnetic inclination, suggests that the Mn-Fe ore of Um Bogma was deposited at a latitude 10° south of the equator, i.e., in tropical conditions. Mineralogical and geochemical investigations support the paleomagnetic results. The ore is enriched with Na, K, Ca, Mg, Sr, Co, and Ni, suggesting that the Mn-Fe ore of Um Bogma is of sedimentary marine origin. The enrichment of some ore samples with hydrothermal elements (As, Zn, Cu, Pb, V, W, and F) is restricted to faulted areas, indicating that a younger phase of hydrothermal activity affected the ore after its deposition.

Samples from the overlying Abu Thora sandstone yield a mean paleomagnetic direction of D = 326°, I = 42°, k = 19.3; α_{95} = 17.9°, which provides a paleomagnetic pole position at latitude 59°, longitude 302° (Dp = 13.5°, Dm = 22°). This magnetization is significantly different from that of the Mn-Fe ore and host rocks and is assigned a Cretaceous age. The porous sandstone acquired its magnetization long after deposition, most likely by diagenetic chemical processes associated with red pigmentation.

Introduction

THE Mn-Fe ore of Um Bogma is situated in southwest Sinai, 20 km east of the Gulf of Suez and 30 km southeast of Abu Zeniema (Fig. 1). Since the discovery of the Mn-Fe ore at Um Bogma by Barron (1907), of the Egyptian Geological Survey, the ore has been subjected to several studies.

Previous geologic studies suggested two origins for the Mn-Fe ore deposit of southwestern Sinai: a hydrothermal, epigenetic origin, and a sedimentary marine origin. The hydrothermal origin for Mn-Fe ore was proposed by Ball (1916); Attia (1956); Weissbrod (1969); Beyth (1987) and Hassan et al. (1989).

Ball (1916) suggested that thermal waters ascending along fault fissures affected the concentration and oxidation of the Mn and Fe carbonates originally contained in the calcareous rocks. However, Attia (1956) and Gill and Ford (1956) considered that the Mn and Fe compounds were carried by hydrothermal fluids that derived from deep felsic igneous rocks. These thermal solutions dissolved the dolomite and deposited the Mn and Fe minerals in its place. On the other hand, El Sharkawi et al. (1990) suggested that the Mn deposit was of supergene origin, accumulated during the mature stage of karstification.

El Shazly et al. (1963), Faris et al. (1963), Mart and Sass (1972), Magaritz and Brenner (1979), Saleeb-Roufaiel et al. (1987), and Zaghloul et al. (1995) have favored the marine sedimentary origin. El Shazly et al. (1963) concluded that ore formation took place in the Carboniferous marine waters at shallow to moderate depths. Iron minerals were concentrated near the shoreline, while Mn minerals tended to be more abundant in deeper water. Both Mn and Fe were derived from weathering of basement rocks. Mart and Sass (1972), however, determined that the Mn-Fe ore had formed during one of the Carboniferous regressions and recognized two stages of the ore formation: during the first, Mn and Fe oxides were formed syngenetically in tidal pools; during the second, cores of the lenses were enriched in Mn while the margins were enriched in Fe.

The commercial production of Um Bogma ore began in 1919 and the total tonnage shipped through 1955 was on the order of 3,300,000 long tons. Since 1956, the production has been nearly 240,000 t per annum of a shipping mixture containing 21.5 percent Mn, 36 percent Fe, and approximately 5 percent SiO_2. Some high-grade MnO_2 is also produced (Gill and Ford, 1956). Kaiser Engineers and Contractors (KECI-ACB; 1980) have calculated that the reserve tonnages from 27 manganese leases granted to Sinai Manganese Co. contain

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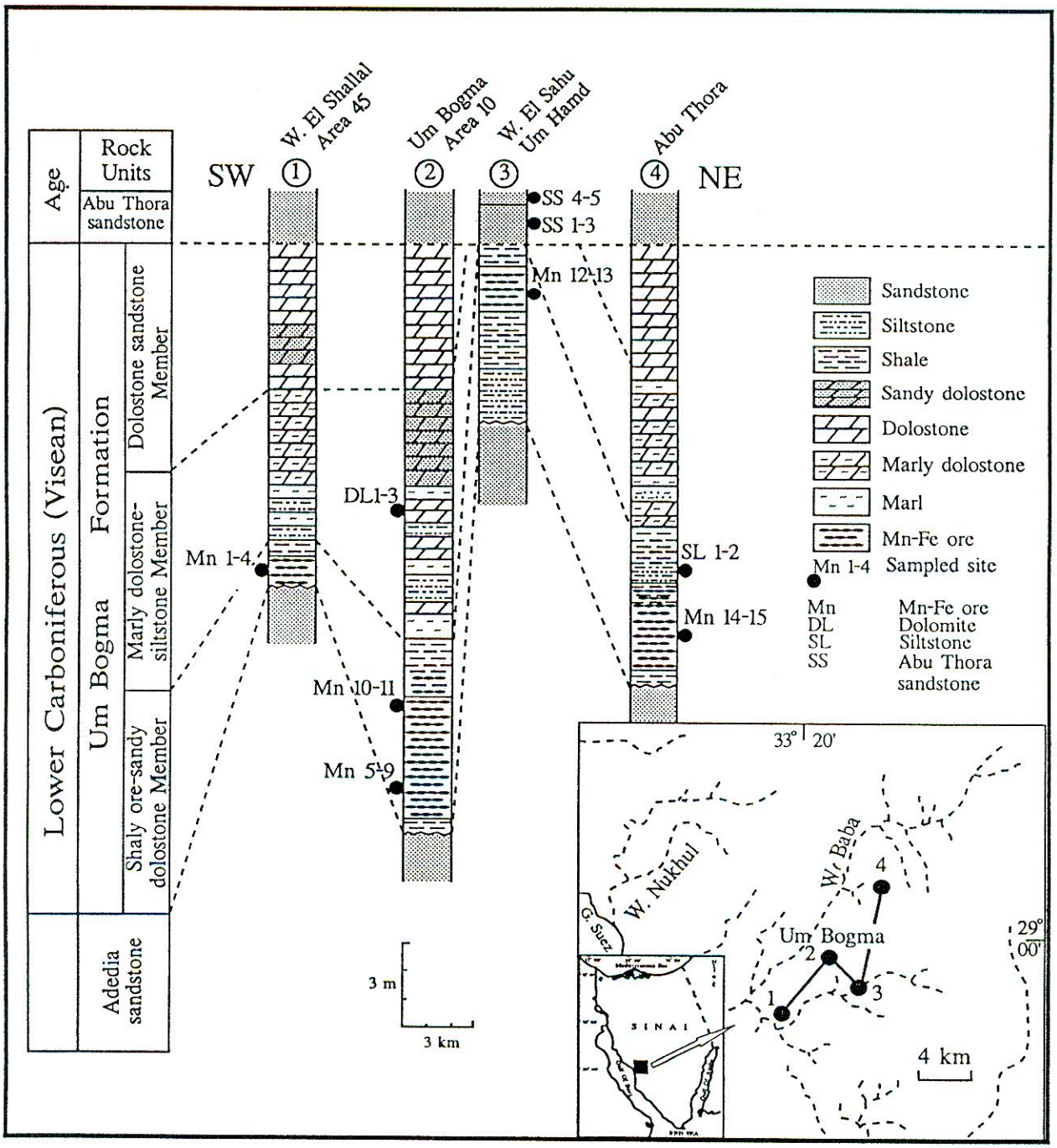


FIG. 1. Stratigraphic correlation chart of measured and sampled sections between Area 45 (western El Shallal) in the southwest and west Abu Thor to northeast of Um Bogma area, southwest Sinai.

1,135,050 t of proven ore at a grade of 27.54 percent Mn and 26.28 percent Fe, with a cumulative Mn/Fe ratio of 1.05:1. Currently, Sinai Ferromanganese Co. produces the ferromanganese alloy by mixing the low-grade ore of Um Bogma with an imported high-grade ore.

Clearly, the age and origin of the Mn-Fe ore deposits of Um Bogma are controversial. The determination of the age of this ore is critical to an understanding of its genesis. Thus, the main objective of the present study is to date these ores using paleomagnetic, geochemical, and mineralogical techniques.

Sampling and Techniques

In the present study the Mn-Fe ore and associated rocks of the Um Bogma Formation and overlying sandstone of Abu Thora Formation (lower Carboniferous) were sampled at four different localities in southwest Sinai (Fig. 1) using a portable motorized drill. Eighty-eight oriented samples were collected from 20 sites in the Mn-Fe ore and associated dolostone and siltstone. In addition, six sites were drilled in the overlying sandstone of the Abu Thora Formation. The sun compass was

used for the orientation of the samples; the magnetic compass was used only in times when the skies were overcast. Figure 1 shows the stratigraphic position of the sampled sites relative to the stratigraphic position of the Mn-Fe ore, with the stratigraphic sequence of the Carboniferous rocks in the sampled localities. The broad geographic distribution of the sampling sites assures that all the Mn-Fe ore deposits are represented in the magnetic remanance measurements.

The mineralogical composition of representative samples was identified using X-ray diffraction technique. The concentrations of major oxides (SiO_2, Al_2O_2, MnO, Fe_2O_3, TiO_2, MgO, CaO, K_2O, Na_2O and P_2O_5) were determined using the methods of Shabiro and Branockis (1962). Trace elements were analyzed using X-ray fluorescence (XRF), inductively coupled plasma (ICP) spectrometry, and inductively coupled plasma-mass spectrometry (ICP-MS); fluorine was determined using the specific ion electrode technique. All geochemical analyses were done in the laboratories of the Nuclear Materials Authority, Egypt and Regina University, Saskatchewan, Canada.

Knowledge of the magnetic mineralogy of the collected samples is necessary to test the reliability of the palaeomagnetic data. Thermomagnetic $M_S(T)$ and hysteresis properties (Day et al., 1977; Parry, 1982) were measured in order to provide some information about the types of magnetic carriers in the samples. natural remanent magnetization (NRM) of the samples was measured on a SQUID cryogenic magnetometer. All the paleomagnetic and magnetic measurements were done in the paleomagnetic laboratory of the University of Liverpool, United Kingdom.

Geologic Setting

The Um Bogma area is located to the east of the Gulf of Suez. This area is dominated by north-northwest- to northwest- and north-south-trending faults (ElAgami, 1996). In the Um Bogma area the Precambrian basement is overlain by Paleozoic sedimentary rocks. The Cambrian succession (Sarabit El Khadim, Abu Hamata, Nasib, and Adedia Formations) comprises continental to shallow marine clastic rocks (see Soliman and El Fetouh, 1969; Kora, 1984; ElAgami, 1996). These rocks are unconformably overlain by Carboniferous shallow marine facies of the Um Bogma Formation and deltaic to fluviomarine facies of the Abu Thora Formation (Fig. 2). Basaltic sills (Late Triassic-early Jurassic) cut the top of the Abu Thora Formation, while northwest-southeast-trending Tertiary doleritic dikes occur throughout the area.

The Um Bogma Formation represents the main target of this study, as most of the mineralization (Mn-Fe ore, copper, U, and gibbsite) occurs within it. The formation is composed mainly of dark gray and pink dolomite, especially in the lower and upper parts, with alternating ocherous yellow marl, ocherous soft earthy dolomite, and soft yellowish siltstones making up the middle section. The formation exhibits a variable thickness; a maximum thickness (41–43 m) has been measured at Wadi Nukhul in the northwestern part of the area of study, decreasing to about 21 m at Um Bogma, with a minimum thickness of 3 m at Gebel Ghorabi. Generally, thickness decreases to the southeast and east.

Ferromanganese ore lenses associated with shaly beds occur at the base of Um Bogma Formation and were exploited at

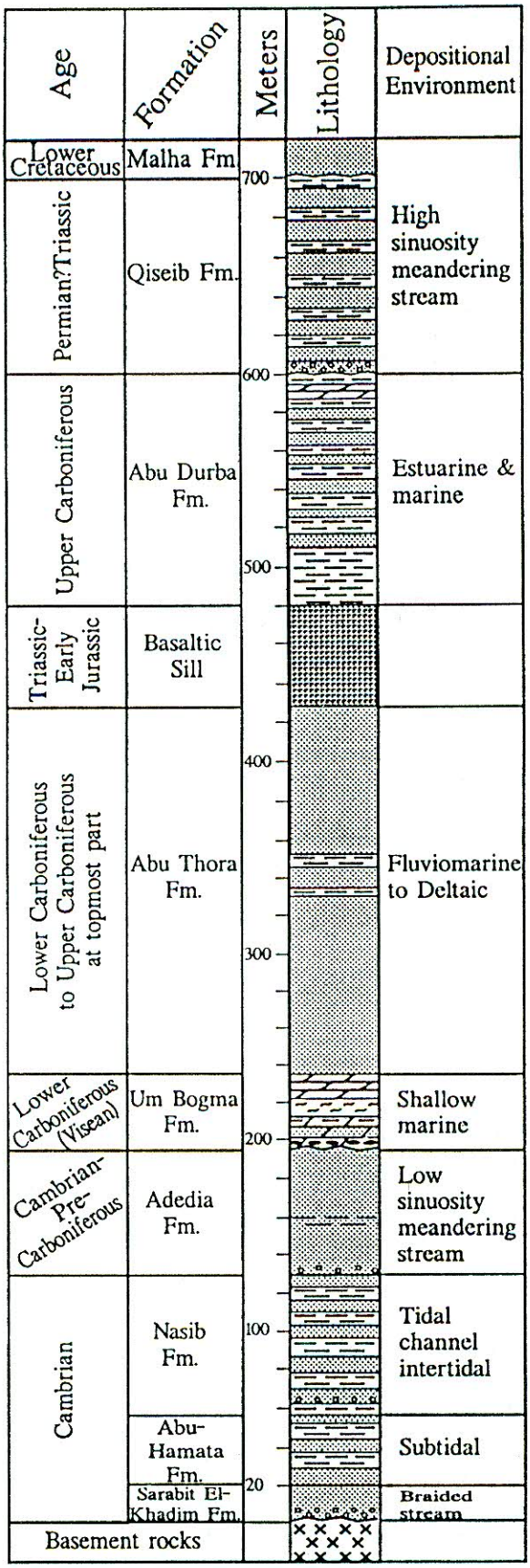


FIG. 2. Stratigraphic subdivisions of Paleozoic rocks in southwest Sinai, Egypt.

Um Bogma, Wadi El Shallal, El Ramsi, Wadi El Kharig, Abu Thora, G. Allouga, Um Hamad, and other localities. An early Carboniferous (Visean) age for the Um Bogma Formation is accepted by most workers, based on micro- and macrofossil contents (Mamet and Omara, 1969; Brenckle and Marchant, 1987; and Kora, 1989).

The Um Bogma Formation has been subdivided by several authors (e.g., Soliman and El Fetouh, 1969; Kora, 1984; Kora et al., 1994). Recently, ElAgami (1996) subdivided this formation into three members: a lower shaly ore-sandy dolostone member, a middle marly dolostone-siltstone member, and an upper dolostone-sandstone member (Fig. 1). The lower shaly ore-sandy dolostone member contains the majority of the Mn-Fe ore. This lower member exhibits three different lithofacies: ferruginous siltstone and silty shale facies, sandy dolostone-gibbsite ore facies, and shale to silty shale facies.

The Mn deposits of Um Bogma occur in the form of tabular and lenticular deposits, more or less regular in form, at or near the base of the dolomitic Um Bogma Formation, and therefore belong to a well-defined time stratigraphic level and show evident lateral facies changes. The Mn bodies are usually surrounded by calcareous shale, siltstone, or sandstone that grades laterally into dolomite. In some localities the Mn-Fe ore overlies the dolomite of the lower member, as recorded at area 45 Wadi El Shallal, Wadi Kharig, El Ramsi, and Abu Thora area. The ore also occurs within dolomite of the upper member at Wadi Khaboba. In the southeast section of the Um Bogma area, the Mn-Fe ore is associated with ferruginous silty shale and directly overlain by the sandstone of the Abu Thora Formation. In this area, the middle and upper members of the Um Bogma Formation are missing.

The Mn orebodies are characterized by an Mn-rich inner core of pyrochroite-hausmannite and an outer core of pislomelane-pyrolusite with high iron content (Mart and Sass, 1972; Magaritz and Brenner, 1979).

Geochemistry

The MnO_2 content of the studied ore deposits ranges from 5 to 53.5 percent, with an average content of 29.6 percent; Fe_2O_3 content ranges from 7.4 to 59.3 percent, with an average of 36 percent. The Fe/Mn ratio ranges from 0.17 to 10.7. Fe is strongly fractionated from Mn within the ore, and the Fe/Mn ratio increases toward the inferred shoreline near Wadi El Sahu, as well as at the periphery of the Mn-Fe ore lenses. The separation of Fe from Mn may be explained by the variation in Eh-pH during diagenesis and deposition (Krauskopf, 1956; Roy, 1981). Such processes could result in the formation of zoned Mn-Fe bodies consisting of Mn-rich cores (high pH and low Eh) and Fe-rich margins (lower pH and high Eh).

Magaritz and Brenner (1979) have suggested that Mn-Fe ore of the Um Bogma area was subjected to various diagenetic processes that resulted in the separation of manganese from iron. Moreover, this separation is a primary depositional one and usually results in near-shore deposits with high Fe/Mn ratios and deep-water deposits with low Fe/Mn ratios. This behavior may account for the iron-rich lenses in the silty facies of the Um Bogma Formation at Wadi El Sahu (Table 1)—i.e., suggesting near-shore deposition—compared with Mn-rich lenses in the deeper-water dolomitic facies (at Wadi El Shallal and Abu Thora).

Mn oxides have a strong sorption capacity for cationic species in the depositional fluids. This sorption capacity is reflected in the trace element composition of the oxides, which differs depending on whether precipitation occurs from fresh water, seawater, or hydrothermal fluids.

Hewett and Fleischer (1960) and Hewett et al. (1963) pioneered the use of trace elements to identify different types of manganese mineralization. They noted that hydrothermal Mn deposits in both terrestrial and marine settings display enrichments in As, B, Ba, Ge, Pb, Sb, Sr, Ti, and W. Subsequent work confirmed that at least some of these elements, together with Li, Cd, Mo, V, and Zn, are consistently enriched in oxides. These oxides derived from hydrothermal fluids in both terrestrial and marine environments (Bostrom and Valdes, 1969; Ossa, 1970; Cronan, 1972; Sillitoe, 1975; Zantop, 1978, 1981; and Varnavas et al., 1988). Recently, Nicholson (1992) postulated that hydrothermal Mn deposits are characterized by enrichment in As, Ba, Cu, Pb, Sb, Sr, Li, Cd, Mo, V, and Zn. Sedimentary marine Mn deposits show Na-K-Ca-Mg-Sr and Co-Cu-Ni enrichment.

Several authors have attempted to differentiate sedimentary marine and hydrothermal Mn deposits using major and trace elements. Bonatti et al. (1972) produced an early version of the well-established Fe-Mn-10(Co + Cu + Ni) ternary plot, which differentiates between sedimentary marine and hydrothermal manganiferous deposits. It has been recognized that hydrothermal oxides are depleted in Co, Cu, Ni, and Zn relative to sedimentary marine deposits. Crerar et al. (1982) recognized hydrothermal deposits using an Si-Al plot. Recently, Nicholson (1992) illustrated the use of an Na(%)-Mg(%) diagnostic plot to differentiate between marine and fresh water sedimentary Mn deposits. In addition, he established the Co + Ni wt percent vs. As + Cu + Mo + Pb + V + Zn wt percent plot to differentiate between the sedimentary marine and hydrothermal manganese oxides.

Since most bivariant and ternary plots were designed to differentiate between different genetic types of Mn deposits, it was reasonable to expect these plots to show whether the Mn ore deposits of the Um Bogma area was sedimentary or hydrothermal in origin. Nicholson plots (1992; Fig. 3A and B) show that Um Bogma samples fall within a marine-fresh water field. One representative sample shows a high content of W (37.6 ppm), F (9,000 ppm), and U (904 ppm); this points to later hydrothermal activity on the fault planes, which caused the enrichment of the ore with hydrothermal diagnostic elements (Cu, Mo, Pb, Zn, and W).

The results of the trace element analysis of Magaritz and Brenner (1979) for Mn-Fe ore of the Um Bogma area were plotted on the same discrimination diagram as the present study (Fig. 3A and B). This plot shows that the Mn-Fe ore samples lie on or near the boundary between marine and hydrothermal deposits (Fig. 3A); they are rich in trace elements characteristic of marine deposits (Co-Ni) as well as in trace elements that characterize the hydrothermal deposits (As, Mo, Cu, V, Zn, Pb). The results suggest that the Mn-Fe ore was deposited in a marine environment, but some areas underwent later hydrothermal alteration that enriched the Mn-Fe material with W, Zn, Pb, Mo, As, and V.

TABLE 1. Chemical Analyses of Major and Trace Elements of Representative Samples of Mn-Fe Ores and Associated Rocks

Major and trace elements	W. El Shallal (Area 45)	Um Bogma (Area 10)		W. El Sahu (Um Hamd)						Abu Thora
	Mn-Fe ore	Mn	Dolom.	Mn-Fe ore				Abu Thora SS		Mn-Fe
Sample no.	Mn1	Mn3	Mn10	DL3	Mn12.2	Mn12.3	Mn13.1	Mn13.3	SS1	SS2	Mn14
Wt percent											
SiO_2	10.90	15.30	3.60	6.60	16.91	5.07	17.39	13.38	65.00	73.50	7.50
Al_2O_3	5.8	6.7	1.1	2.4	9.91	1.98	5.35	4.96	2.5	8.4	0.6
Fe_2O_3	7.4	53.5	38.1	2.3	34.63	37.61	27.32	59.3	21.5	1.75	30.1
MnO	42.5	5.0	33.5	0.0	23.79	42.19	23.7	11.9	0.4	0.15	54.3
Fe/Mn	0.17	10.7	1.1		1.46	0.89	1.15	4.98			0.55
MgO	1.6	1.5	1.21	18.2	1.1	0.88	0.9	0.85	3.0	5.0	2.1
Na_2O	0.08	0.61	0.7		0.75	0.69	1.33	0.65	0.06	1.54	0.85
K_2O	0.12	1.34	0.21		0.85	0.45	0.29	0.24	0.04	1.08	0.07
TiO_2	0.25	0.24	0.0		0.22	0.08	0.35	0.18	0.07	0.83	0.20
P_2O_5	0.5	0.3	0.15		1.62	0.29	2.43	0.61	0.8	0.46	0.09
LOI	14.98	10.5	9.10	42.5	8.99	10.5	11.57	8.18	0.98	3.55	2.1
Total	94.03	98.81	95.93	100.0	98.06	100.0	93.08	100.5	99.37	99.07	100.0
Parts per million											
As	320	210	132		1,220	170	278	130			240
Ba	910	2,400	1,500	550	237	7,653	15,833	626	820	267	900
Co	213	310	260	283	238.3	412.2	182.7	241.5			210
Cr		65	55	67	180	69	310	70	40	34	45
Cu	410	320	1,450	280	964	746	728	610	10	14	440
Mo	6.3	8.3	7.6		10.6	7.1	7.4	8.2			6.5
Nb	12.0	14.0	11.0	29.0	11.0	4.2	15.9	29	4.0	44.0	14.5
Ni	200	500	240	128	463	393	265	192	540		250
Pb	220	70	450	11	371	27	922	180	33	520	385
Rb	4.0	3.0	5.0		11.8	13.9	8.7	2.5			2.0
Sr	530	81	130	120	575	428	2267	813	49	247	410
Th	3.0	9.0	9.0	2.0	9.9	4.2	11.6	66.0	14.0	15.0	2.0
U	0.5	1.0	0.5	1.0	904	197.3	660.7	200	117	146	0.5
U/Th	0.17	0.11	0.06	0.5	91.3	46.98	56.96	3.03	8.36	9.73	0.25
Tl	0.04	0.02	0.05		0.1	0.05	0.2	0.1			
V	240	270	250	159	384	142	457	210	260	320	320
W	1.0	2.0	3.0			37.6	2.3	33.1	2.0		
Zn	380	620	410	64	2,303	598	1,982	308	72	65	430
F	Nd	Nd	Nd	Nd	9,000	Nd	Nd	Nd	Nd	Nd	Nd

The concentrations of major oxides were determined using the method of Shabiro and Branockis (1962); trace elements were analyzed using XRF, ICP, ICP-MS; fluorine was determined using Specific Ion Electrode technique

LOI = loss on ignition; Nd = no data

The results of the present study, together with those of Magaritz and Brenner (1979), were plotted on a ternary plot (Fig. 4) of Bonatti et al. (1972) and Crerar et al. (1982). These plots show that Mn-Fe ore of the Um Bogma area lies on the boundary lines between hydrothermal and sedimentary Mn deposits, indicating that these deposits have geochemical signatures indicative of both marine and hydrothermal origin.

For further confirmation of the genesis of Um Bogma ore, the results of the present work were compiled with results of Zaghloul et al. (1995) and ElAgami (1996) and plotted on a bivariant plot (Si vs. Al wt %; Fig. 5) of Crerar et al. (1982). This shows that the studied Mn-Fe ore deposits are of sedimentary-detrital origin (marine). Also, the present data were compiled with the results of ElAgami (1996) and plotted on the diagnostic plot of Nicholson (1992; Fig. 6) to differentiate between marine and fresh water deposits. Results indicate that the studied Mn-Fe deposits are of marine environment.

The statistical analysis for the geochemical results of Um Bogma Mn-Fe ore of Magaritz and Brenner (1979) and ElAgami (1996) shows positive correlation with Co, Ni, and Cu and strongly correlates with Sr, Mg, and Na. According to the scheme of Nicholson (1992), such positive correlation points to a marine environment.

The U/Th ratio may similarly be used to discriminate hydrothermal deposits from sediments and normal pelagic sediments (Bonatti, 1975). The Mn-rich ore of the Um Bogma area (Wadi El Shallal, Area 10, and Abu Thora) has a low U/Th ratio (0.06–0.17), which is in the range of typical marine sediments, where the sediments are sufficiently and slowly accumulated to incorporate significant Th from seawater. On the other hand, the U/Th ratio for Fe-rich ore and ore associated with terrigenous materials or near structural elements (Wadi El Sahu, Um Hamd) is high (3.03 to 91.3), several orders of magnitude greater than typical marine sediments, indicating a hydrothermal origin for these deposits. These U/Th ratios provide firm evidence for a later hydrothermal enrichment of the Mn-Fe ore after its primary deposition in a marine environment.

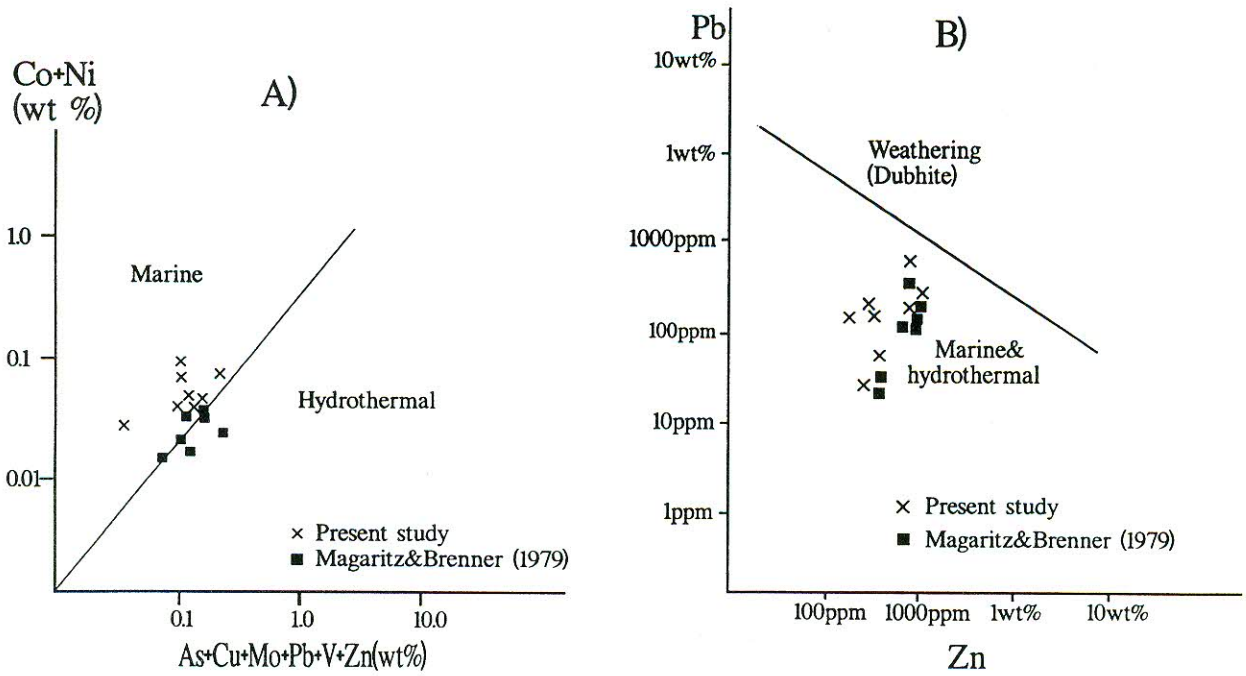


FIG. 3. Diagnostic plots to differentiate sedimentary marine and hydrothermal manganese oxide (A) and weathering (dubhites) from other sedimentary marine and hydrothermal oxides (B). Dubhites are Mn oxides drived from weathering of mineralized sequence (Nicholson, 1992).

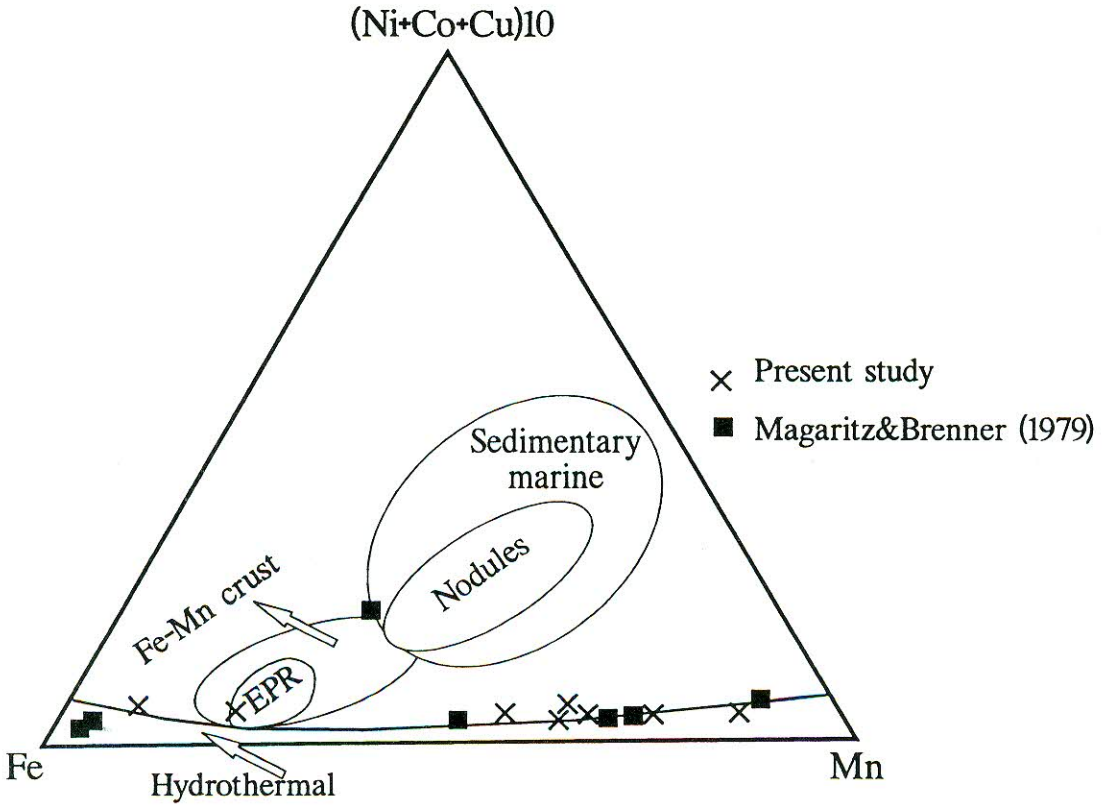


FIG. 4. Ternary diagram shows Fe vs. Mn vs. (Ni + Co + Cu)10 for manganiferous marine deposits. The diagram compares data from Mn-Fe ore lenses of Um Bogma deposits with generalized fields for marine and hydrothermal deposits (after Bonatti et al., 1972). Also, the diagram shows ore fields of the East Pacific Rise (EPR) metalliferous sediments (after Corliss and Dymond, 1975) and fields of intermediate Fe-Mn crusts ad ferromanganese nodules taken from Toth (1980).

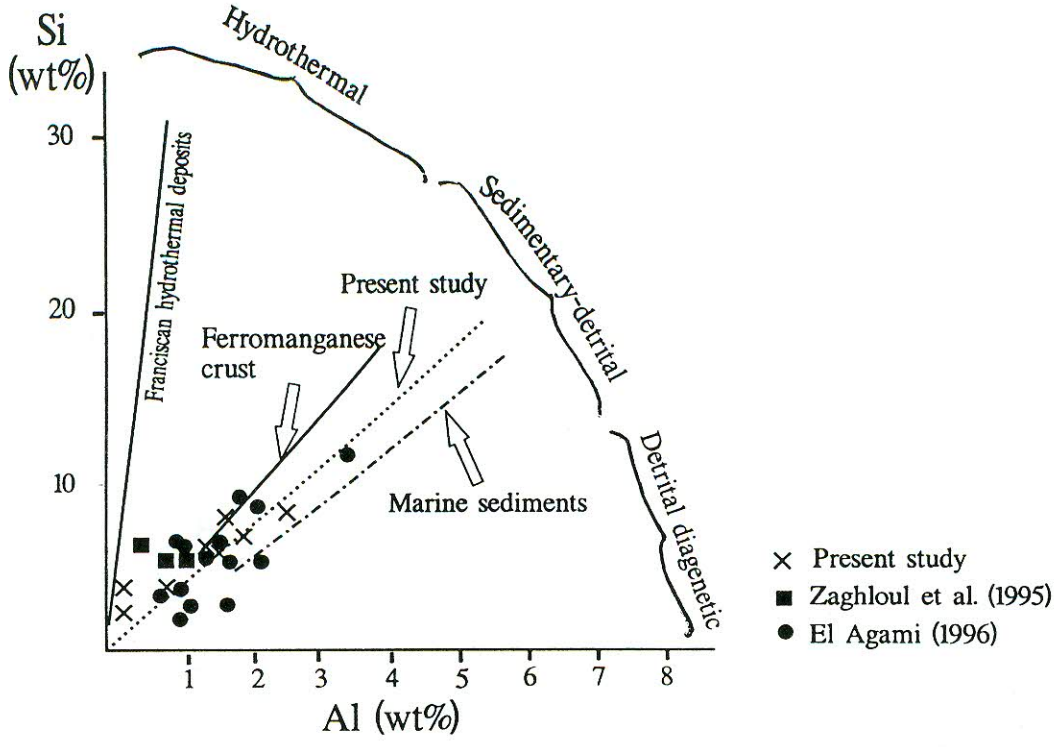


FIG. 5. Si vs. Al, comparing analyses of Um Bogma Mn-Fe ore with fields of typical marine sediments. Data for modern Mn nodules are from Calvert and Price (1977) and Toth (1980); modern hydrothermal Fe-Mn deposits from Bonatti (1975) and Toth (1980); ferromanganese crusts from Toth (1980); average marine sediments from Turekian and Wedepohl (1961). After Crerar et al., 1982.

Mineralogy

The mineralogy of the Mn deposits is (from most abundant to least abundant mineral species): pyrolusite, psilomelane (romanèchite)-cryptomelane, manganite, δ Mn oxide, Mn oxide hydroxide iron [$(Fe_{67}Mn_{33})OOH$], coronadite, hollandite, hetaerolite, hydrohetaerolite, $ZnMnO_3$, and bixbyite. Mn minerals of high oxidation state (pyrolusite and psilomelane) are the most dominant, while those of low oxidation state are only rare constituents. Goethite and hematite are intergrown with Mn minerals in the majority of studied ore samples. Gangue minerals include quartz, dolomite, barite, calcite, gypsum, clay, gibbsite, and alunite.

Mineralogy as a tool for characterizing the genetic origin of Mn deposits has been used occasionally, but a diagnostic mineralogical signature has not been rigorously defined. Recently, Nicholson (1992) concluded that bixbyite, braunite, hausmannite, hubnerite, jacobsite, and pyrochroite are of dominantly hydrothermal origin, whereas chalcophanite, coronadite, crednerite, δ-MnO_2, groutite, hollandite, lithiophorite, manganite, nsutite, quenselite, ramsdellite, todorokite, and woodruffite are typical in sedimentary deposits. Romanèchite is also predominantly sedimentary in origin and rarely found in hydrothermal deposits.

In this context the mineral-forming associations of the studied deposits are typical for sedimentary Mn deposits (marine deposits). The bixbyite is only recorded in Wadi El Sahu ore samples, suggesting a secondary hydrothermal signature, which is confirmed by the geochemical results.

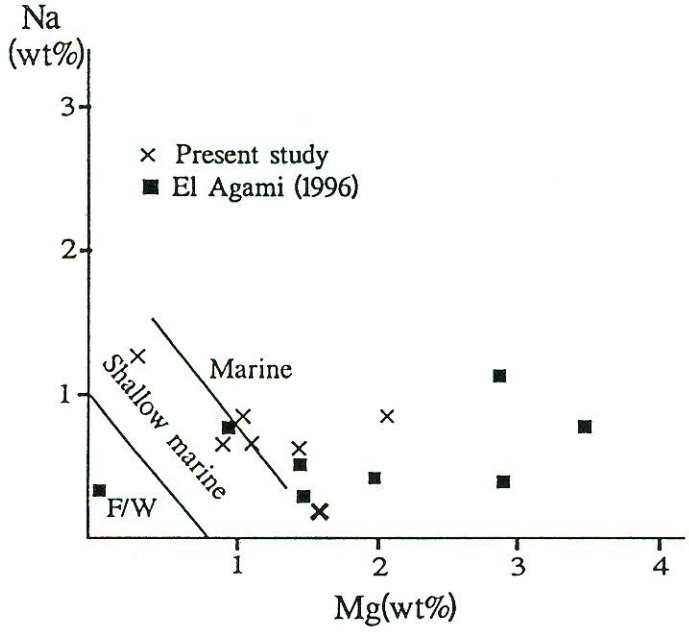


FIG. 6. Diagnostic plot to differentiate marine and fresh water (F/W) deposits (after Nicholson, 1992).

Magnetic Mineralogy

Representative specimens from the sampled sites were used for measuring the thermomagnetic characteristics (normalized saturation magnetization against temperature), in order to determine their behavior during heating and cooling, and to reveal the characteristic Curie temperature of the principal magnetic minerals. Figure 7A, C, and E shows representative examples of thermomagnetic curves $M_s(T)$ of different samples from Mn-Fe ore, associated rocks, and overlying sandstone. All $M_s(T)$ curves show T_C at ≅680°C, which is typical for hematite. In addition to hematite, some sandstone samples show a Curie temperature, T_C,of about 120°C, which is representative of goethite (Fig. 7C). Sample Mn-Fe 6.3 (Fig. 7A) shows a change in the M_S (T) curve at ≅270°C, which may suggest the presence of a very small amount of siderite in addition to hematite. The presence of siderite in the Um Bogma Mn-Fe ore was noted by Anwar and El Sokkary (1970).

Hysteresis loops at room temperature for representative samples from the different sampled ore and rocks were obtained by using the vibrating sample magnetometers (VSM) technique, and results are shown in Figure 7 (B, D, F). These loops indicate the presence of a high coercive hematite mineral in the form of single domain (Fig. 7D) and/or multidomain in character (Fig. 7F).

Paleomagnetic Investigations

Paleomagnetic measurements

Paleomagnetism of sedimentary rocks has significance in determining the age of deposition and diagenetic processes

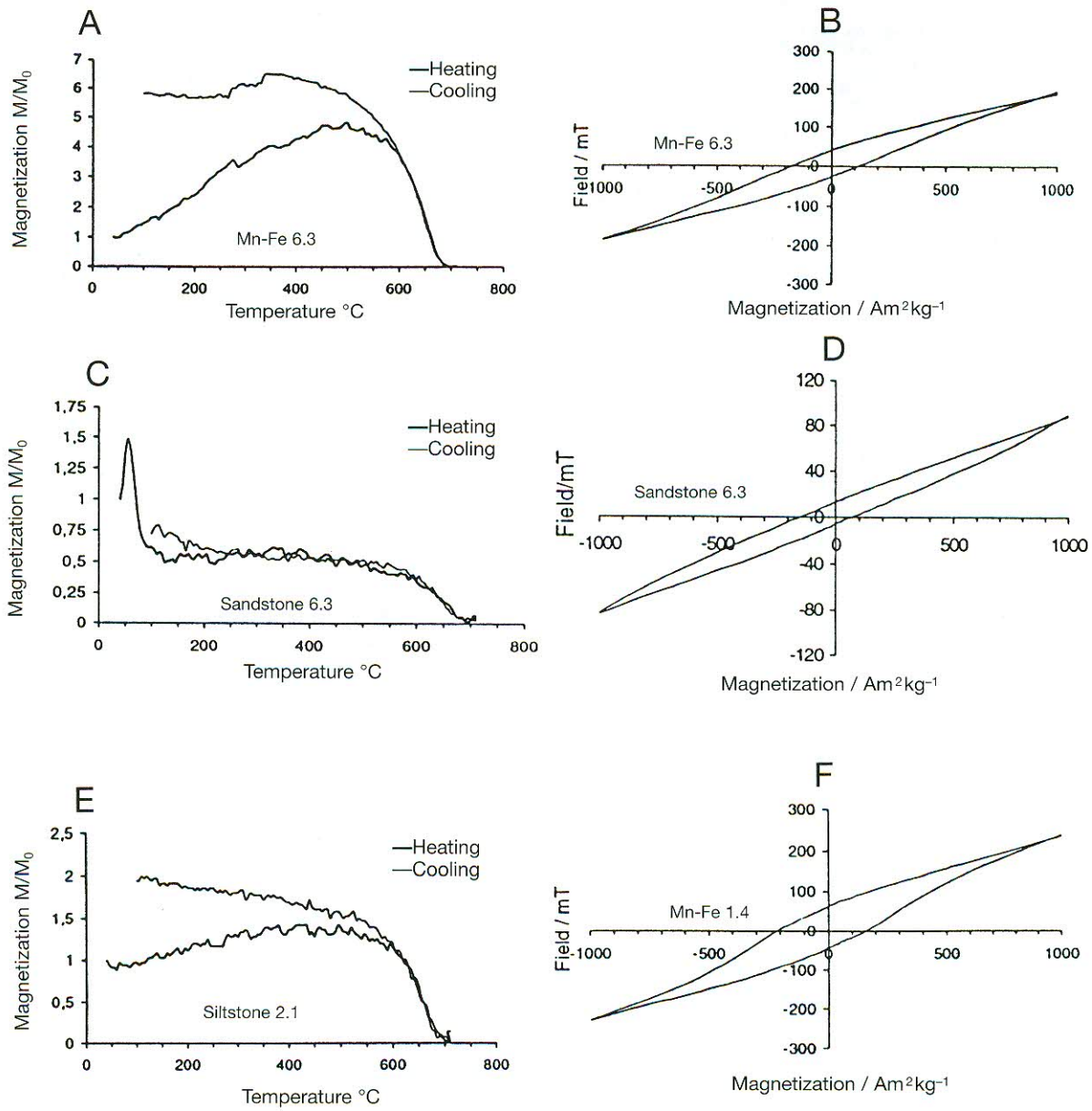


FIG. 7. Representative examples for the thermomagnetic curves (A,C, E) and hysteresis loops (B, D, F) of the studied rock types.

that have affected the rock since its formation (see Ibrahim et al., 1998). The NRM of the samples was initially measured using a Digico spinner magnetometer. It was found that most of the samples, especially the Mn samples, have low intensities (ranging from 0.049 to 3.7 mA/m); thus, all samples were measured on the SQUID cryogenic magnetometer.

Alternating field (AF) demagnetization was applied on one sample from each site, up to 150 mT. Throughout the AF demagnetization the samples showed no significant directional changes. Consequently, all samples were subjected to a full range of thermal demagnetization, using a Shaw Furnace, in steps up to 680°C. The results of thermal demagnetization indicate the presence of high blocking temperature components except in some sandstone samples, which showed a low blocking temperature component—possibly carried by goethite—as well as the high blocking temperature component. This high blocking temperature and high coercivity reveal that hematite is the principal magnetic carrier in the samples. Typical examples for Zijderveld demagnetization plots (Fig. 8) show that characteristic remanent magnetization end-point directions could be identified for most of the samples. The principal component method of Kirschvink (1980) was used to determine these stable directions.

Paleomagnetic results

The paleomagnetic results will be discussed as two separate groups, one including Mn-Fe ore and associated rocks of the Um Bogma Formation, the other including sandstone of Abu

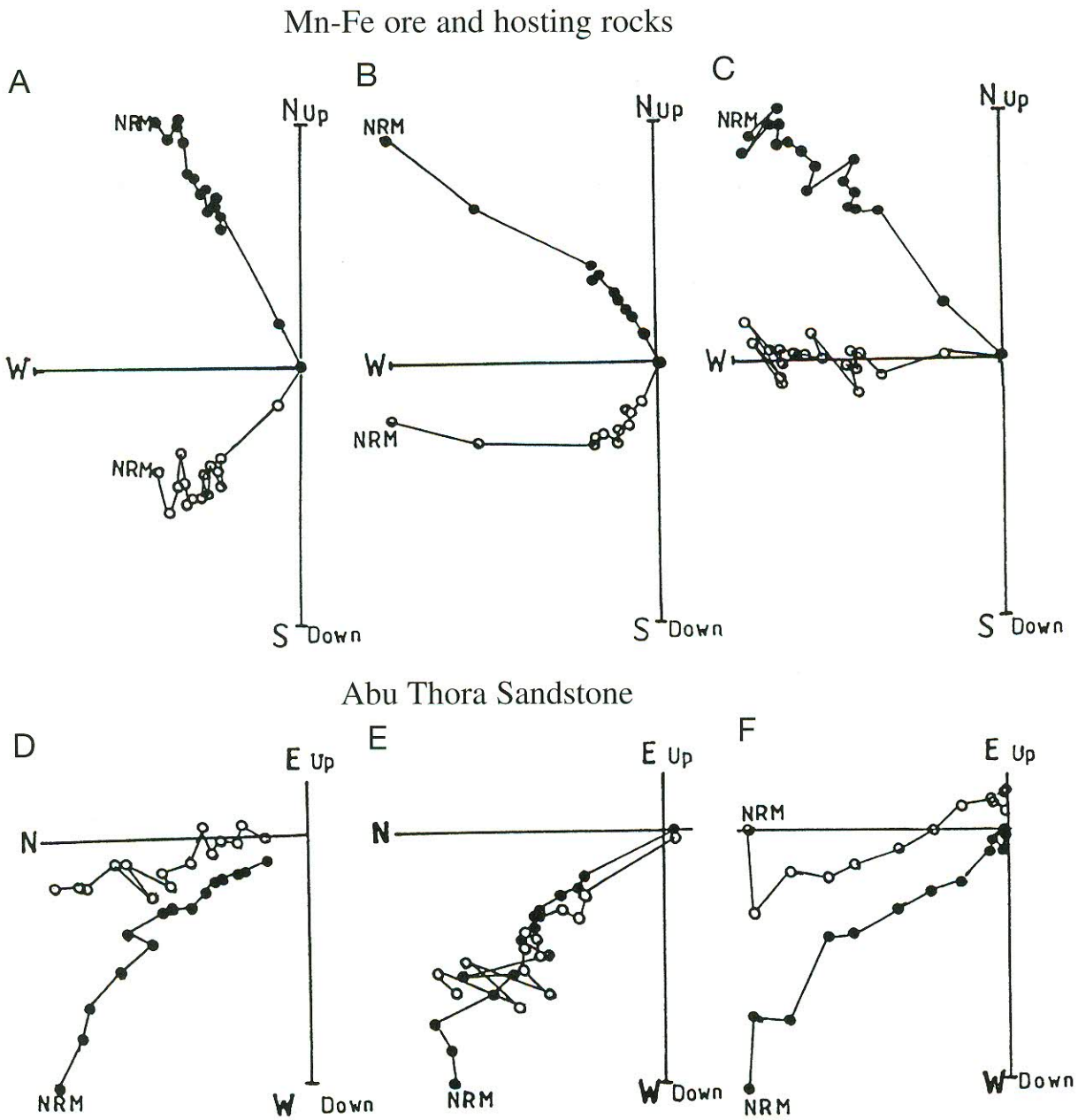


FIG. 8. Typical examples of Zijderveld demagnetization diagrams of the studied rock samples.

Thora Formation. These paleomagnetic results with their statistical parameters are summarized in Table 2.

Mn-Fe ore and associated rocks (Um Bogma Formation): The initial intensities of the NRM of Mn-Fe ore samples vary from 0.049 to 3.7 mA/m. In general, the intensities were low for samples obtained from the center of the ore lenses but higher for those obtained from the periphery because the cores of the lenses are enriched in Mn, whereas the margins are enriched in Fe.

In spite of the application of thermal demagnetization, there is a considerable within-site scatter ($\alpha_{95} > 10°$) in most of the sites. This scatter can be explained in terms of imperfect alignment of magnetic grains during the deposition process, and/or the NRM being acquired in a directionally variable field. Error in the azimuth orientation during field sampling is not excluded. The within-site scatter is not significant at the level of a single site because each site direction can be treated as an independent reading of the ancient geomagnetic field (McFadden, 1982). Thus, a realistic estimate of the mean direction of the magnetization of a rock unit is figured from the average of the directions of the all individual sites that came from this rock unit.

The sample directions were used for site mean directions following the method of Fisher (1953) and converted to virtual geomagnetic poles (VGP; see Table 2). Sites Mn4, Mn9 and Mn15 were rejected because they have scattered NRM that changed erratically during thermal demagnetization. Giving unit weight to the remaining sites and reversing the opposite magnetic directions, the overall mean direction of magnetization is D = 306°, I = –21°; α_{95} = 8.8°. This mean direction yields a paleopole position at 24° N, 274° E (Dp = 4.9°, Dm = 9.3°). The site mean directions and the overall mean direction of magnetization with its α_{95} circle of confidence are plotted in Figure 9A.

Five sites that were drilled in the Mn-Fe ore associated rocks of the Um Bogma Formation, namely dolomite and siltstone, showed paleomagnetic directions with initial intensities ranging from 0.01 to 5.01 mA/m. The site mean directions of magnetization have an overall direction of magnetization of D = 297°, I = –13°; α_{95} = 12.7° which yields a paleomagnetic pole position at 20° N, 283° E (Dp = 6.6°, Dm = 12.9°). These paleomagnetic results show a relatively high between-site dispersion (α_{95} = 12.7°) which may be due to the fact that this value is calculated from only five sites. Figure 9B shows the site mean directions and the overall mean direction of magnetization with its 95 percenat circle of confidence.

Abu Thora sandstone: The initial intensity of magnetization for the studied sandstone samples ranges from 0.025 to 2.4

TABLE 2. Paleomagnetic Data and Associated Statistical Parameters

Site	Site mean directions					Virtual geomagnetic pole positions			
	Dec. (°)	Inc. (°)	N (%)	α_{95} (°)	k (°)	Lat. °N	Long. °E	Dp	Dm
Mn-Fe ore and associated rocks (Um Bogma Formation)									
Mn–Fe 1	296	–17	5	12.2	40.2	17.8	282.3	6.5	12.6
Mn–Fe 2	309	–17	3	20	37.0	28.1	273.9	11	21
Mn–Fe 3	119	19	5	10.3	56	–19.6	99.5	5.6	10.7
Mn–Fe 4°	Unstable and scattered directions								
Mn–Fe 5	115	22	4	8.2	127.9	–15.4	100.5	4.6	8.7
Mn–Fe 6	112	18	6	6.9	94.2	–14.5	104.1	3.7	7.2
Mn–Fe 7	129	22	5	17.4	20.3	–26.3	91.5	9.7	18.4
Mn–Fe 8	307	–24	5	14.6	28.4	24.1	272.0	8.3	15.6
Mn–Fe 9°	Unstable and scattered directions								
Mn–Fe 10	296	–18	5	12.1	41.0	17.5	281.8	6.5	12.6
Mn–Fe 11	326	–24	3	3	523	37.0	265.5	1.9	3.4
Mn–Fe 12°	304	–32	6	32.1	15.8	18.8	270.	18.3	34.3
Mn–Fe 13	340	–26.6	5	8	96	42.8	240.2	4.7	8.7
Mn-Fe 14°	313	–55	4	31.0	9.7	11.8	250.8	31	44
Mn-Fe 15°	Unstable and scattered directions								
Mean	306	–21	10 sites	8.8	31.2	–24	94	4.9	9.3
Siltstone 1	120.9	29.9	3	12.4	64	–17	93.0	7.6	13.7
Siltstone 2	108.2	7.6	4	14.1	25	–14	111.0	7.1	14.2
Dolomite 1	287	–9	5	26	9.6	12.5	290.9	13.2	26.6
Dolomite 2	308	–11	3	19.9	39.5	29.3	277.4	10.2	20.2
Dolomite 3	304	–8	5	11.8	43.3	27.0	281.5	6	11.9
Mean	297	–13	5 sites	12.7	37.5	20.0	283	6.6	12.9
Overall mean	303.0	–19.0	15 sites	7.5	30.1	–22	97	3.9	7.4
Sandstone (Abu Thora Formation)									
Sandstone 1	333	26	6	8.2	55	60.7	277.6	4.8	8.9
Sandstone 2	Unstable and scattered directions								
Sandstone 3	340	39	7	9.1	45	70.7	286.8	6.5	10.9
Sandstone 4	333	42	4	15.8	35	65.5	298.5	11.9	19.4
Sandstone 5	322	38	6	12.7	28.9	54.9	299.6	8.9	15.0
Sandstone 6	285.1	58.7	3	14.1	76	28.9	334.9	15.7	21.1
Mean	326	42.0	5 sites	17.9	19.3	59.0	302.0	13.5	22.0

Abbreviations: Dec. = declination, Inc. ° = inclination, α_{95} = 95% circle of confidence, k = estimation of precision parameter (Fisher, 1953), Lat. °N = latitude (positive north), Long. ° E = longitude (positive east), Dp and Dm = paleopole confidence limits, ° = excluded sites

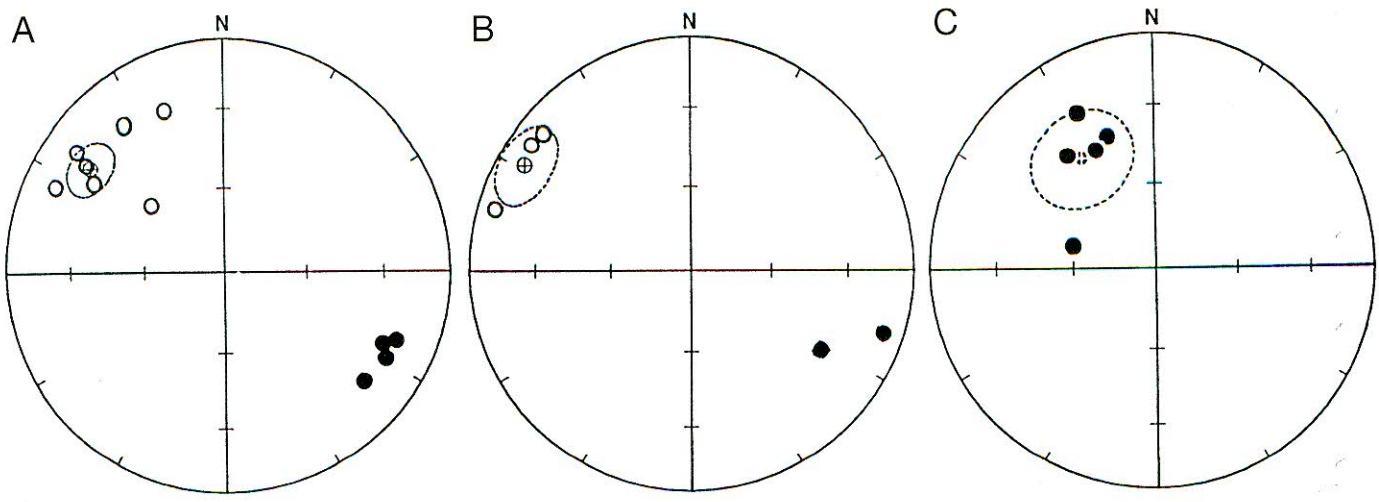


FIG. 9. Site mean directions of magnetization and the overall mean direction of magnetization with its circle of confidence for the Mn-Fe ore (A), associated rocks (B) and Abu Thora sandstone (C).

mA/m. The intensity of magnetization for Abu Thora sandstone is directly related to the degree of red pigmentation; i.e., the redder the sample, the higher its magnetic intensity.

After thermal demagnetization, the within-site scatter as well as between-site scatter was reduced. All sites have normal polarity with moderate positive inclinations and all site directions are considered consistent (Table 2; Fig. 9C). Assigning unit weight to each site, the mean direction of magnetization is D = 326°, I = 42°; α_{95} = 17.9°, which yields a paleomagnetic pole position at 59° N; 302° E (Dp = 13.5°, Dm = 22°).

Discussion and Conclusion

Mineralogical investigations revealed that the Mn-Fe ore and associated rocks of Um Bogma Formation retain a primary magnetization that resides dominantly in single-domain and multidomain hematite that constitute the main magnetic mineral in the studied samples.

Comparatively, the paleomagnetic pole of the dolomite and siltstone of the Um Bogma Formation is different from that of the sandstone of Abu Thora Formation, but there is no significant difference between the Um Bogma pole and the pole of the Mn-Fe ore. Furthermore, the sampled sites from the Mn-Fe ore and those of the associated dolomite and siltstone are of the same stratigraphic age, so their site mean directions are combined in order to calculate a mean paleomagnetic direction, which represents the whole formation. This yields an overall mean direction of magnetization of D = 303°, I = –19°; α_{95} = 7.5°, and a corresponding paleomagnetic pole at 22° S, 97° E (Dp = 3.9°, Dm = 7.4°). Comparison of the observed paleomagnetic poles from the Um Bogma Formation (Mn-Fe ore and host rocks) with the available Carboniferous paleopoles from Africa (Fig. 10) indicates that there is no significant latitudinal displacement, but there is a small longitudinal shift, which could be explained by a local tectonic rotation as the result of rigid block deformation in this area at the margin of the Gulf of Suez. The position of the paleomagnetic pole from Mn-Fe ore and associated rocks close to the reliable Carboniferous paleomagnetic poles from Africa indicates that the ore deposits and the associated rocks of the Um Bogma Formation retain a primary magnetization acquired during the deposition in the Carboniferous. These paleomagnetic results support the sedimentary origin of the Mn-Fe ore, which is consistent with the model proposed by Zaghloul et al. (1995).

The field investigations reveal that the ore lenses, in most cases, are intimately associated with a fine detrital facies (siltstone and shale) underlying, overlying, or lying laterally within the orebody. The paleomagnetic data of these detrital sediments yield the same Carboniferous age as the Mn-Fe ore lenses, suggesting that these sediments may have accompanied the ore during its transportation from land weathering surface to be deposited together in the depression on the floor of the invading shallow sea.

Based on the paleomagnetic direction of Mn-Fe ore and associated rocks, the paleolatitude of the studied locality of Sinai was at a latitude 10° south of the equator during the deposition. According to this paleolatitude and the classification of Brooks (1949), Sinai as an incorporated part of Africa was situated in a tropical climate. This is confirmed by the conclusions of Kora (1989) and El Shahat and Kora (1988), who noted that the early Carboniferous fauna of Um Bogma area have subtropical characteristics. These climate conditions helped in the formation of Mn-Fe ore, as long, wet tropical weathering allowed Mn to migrate away from the weathering surface and precipitate as tetravalent manganese oxides of high Mn/Fe ratio. Concentration of Mn relative to Fe is commonly a phenomenon of a wet, tropical climate (Crerar et al., 1972). Moreover, the tropical climate during the Carboniferous was reflected in the formation of gibbsite associated with some Mn-Fe ore.

There is a clear lateral facies change of the Um Bogma Formation. Toward the east and southeast, near-shore deposition was predominantly detrital and the iron predominates (e.g., Wadi El Sahu, Um Hamad). In the west and northwest, on the other hand, relatively deep water deposition resulted in a thicker succession of dolomite associated with stratiform Mn ore (Wadi El Shallal and Abu Thor), and Mn predominates.

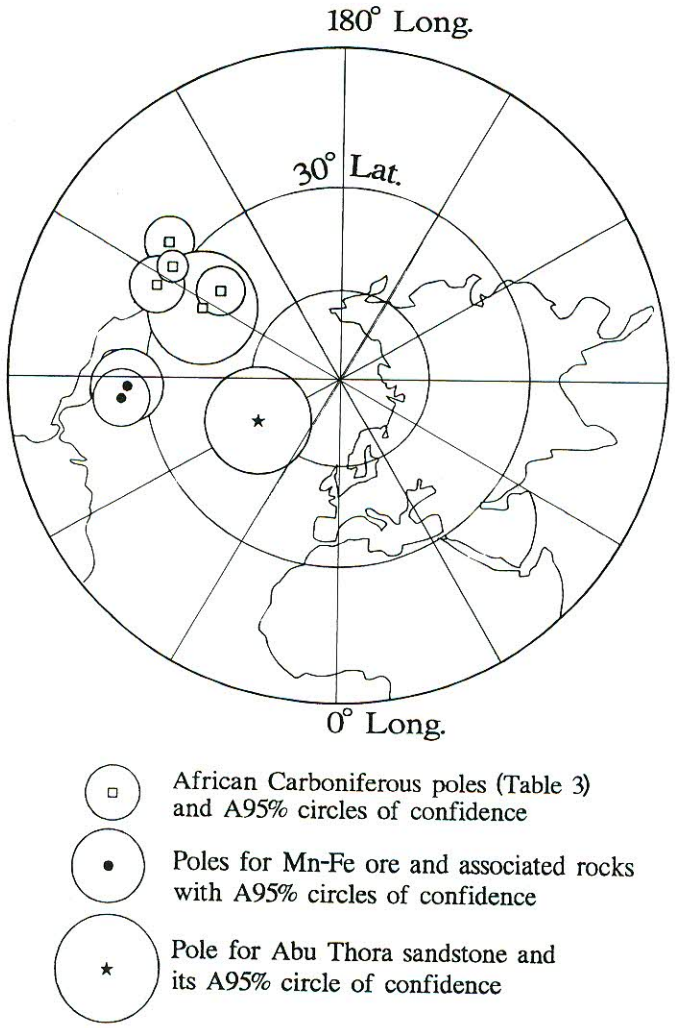


FIG. 10. Comparison between paleomagnetic pole positions derived from rocks sampled in the present study and reliable Carboniferous paleomagnetic poles for stable Africa (Table 3).

These facies changes are reflected in the magnetic intensities of the samples, in which the intensity of magnetization increases with increasing the amount of iron present in the samples.

The field investigation indicated that the Mn-Fe ore deposits are older than the predominant faulting in the area, e.g., Area 10 Um Bogma where the Mn-Fe ore is displaced by Oligo-Miocene faults and cut by dolerite dikes. This is confirmed by the paleomagnetic results, which indicate that the Mn-Fe ore and associated rocks at this locality are of Carboniferous age, i.e., older than the Oligo-Miocene faulting and dikes.

Characteristic mineralogical-geochemical enrichment and geochemical association signature, in conjunction with several diagnostic plots of Crerar et al. (1982) and Nicholson (1992), indicate that the ore is sedimentary in origin. The ore is enriched in Na, K, Ca, Mg, Sr, Co, Cu, and Ni. Other samples have the criteria of both marine and hydrothermal deposits; they are enriched with Co, Ni as well as Cu, As, Zn, Pb, V, W, and F. These samples were restricted to the faulted areas, where the ore was enriched with such hydrothermal elements during the younger mineralization phases associated with the rifting of the Gulf of Suez.

The paleomagnetic results from sandstone of Abu Thora Formation show a paleomagnetic direction that is significantly different from that of Mn-Fe ore and the associated rocks of the Um Bogma Formation. Comparison of the paleomagnetic direction from the sandstone with reliable African Carboniferous poles indicates that the age of the magnetization of the sandstone must be younger than its stratigraphic age (Fig. 10). It is believed that these sediments acquired a secondary magnetization that is related to red hematitic pigmentation. This magnetization was more probably acquired by diagenetic chemical processes that affected the rock after its formation.

An alternative explanation is that the observed remanence was acquired quickly, long after the formation of the rock, and that secondary magnetization obscured the initial primary remanence. In assigning the age of the closest point on the polar wander path of Africa to the remagnetization pole position, a Cretaceous age is suggested. Moreover, the normal polarity of the sandstone magnetization is consistent with the long period of quiet during the Cretaceous, when normal polarity lasted from Early Aptian to the Santonian-Campanian (Lowrie and Alvarez, 1977; Channell et al., 1979). It is believed that the hematite pigments that stained this sedimentary rock could have been formed at low temperature by the dehydration of goethite (Van Houten, 1961, 1968) and/or under the effects of hydrothermal solutions.

Based on all these paleomagnetic, mineralogical, and geochemical results, the authors propose a sedimentary origin of the Mn-Fe ore deposits of Um Bogma.

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September 29, 1998; September 25, 1999

TABLE 3. Reliable Carboniferous Paleomagnetic Pole Positions from Africa

Rock units	Age	Lat. °N	Long. °E	α_{95} (°)	Reference
Dwyka varves	Cu	26	206	10	McElhinny and Opdyke (1968)
Sabaloka comples, Sudan	400 Ma	42	236	7	Soffel et al. (1990)
Ain-Ech Chebbi RB, Algeria	Cu	23	232	6	Daly and Irving (1983)
Hassi-Bechar RB, Algeria	Cu	27	237	3.7	Daly and Irving (1983)
Um Bogma Fm, SW Sinai	Cl	22	277	7.5	Present study
Abu Durba sediments, Egypt	Cu	26	244	7.1	Abdeldayem et al. (1994)

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